DATA-DRIVEN DECENTRALIZED SUPERVISORY VOLT/WATT CONTROL IN PV-RICH ACTIVE DISTRIBUTION NETWORKS

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Abstract
High PV penetration in low-voltage feeders can push bus voltages beyond secure limits. Conventional mitigation uses fixed Volt/Watt droop curves tuned offline, which can underperform as operating conditions vary. We propose a Multi-Agent Reinforcement Learning (MARL) controller that adjusts PV active-power setpoints online based on current operating conditions to maintain voltages, maximize PV utilization, and promote fairness among units. A shared policy is trained on diverse PV/load profiles with a reward penalizing voltage violations, curtailed energy, and availability-weighted curtailment disparity. A Deep Neural Network (DNN)-based safety layer filters potentially unsafe actions using a learned approximation of a voltage-feasibility projection. On a realistic residential feeder with measured-style time series, the proposed method outperforms tuned Volt/Watt droop and a sensitivity-based equal-curtailment baseline under quasi-static operating conditions, reducing violation magnitude and duration, increasing injected PV energy, and equalizing curtailment without per-scenario retuning. These results indicate that learning-based control can deliver adaptive and fair supervisory Volt/Watt behavior in PV-rich low-voltage feeders.

1 Introduction

Rapid growth of rooftop photovoltaics (PV) and other Distributed Energy Resources (DERs) is transforming passive feeders into Active Distribution Networks (ADNs), where instantaneous DER output can exceed local demand and drive bus voltages above statutory limits [1]. To prevent over-voltage, utilities increasingly rely on Volt/Watt Control (VWC) implemented in smart inverters: as the measured voltage rises, the inverter curtails active power, limiting reverse power flow and avoiding nuisance trips [2]. Although Volt/Var control can mitigate both over- and under-voltage without curtailing generation, its performance degrades in low-voltage feeders with high R/X ratios and limited reactive power capability [3]. VWC is attractive because it is simple, local, and inexpensive compared with network reinforcement, and active-power control capabilities are specified for DERs in interconnection standards such as IEEE Std 1547-2018 [4].

However, fixed, standardized Volt/Watt (VW) curves pose important challenges. Poorly tuned settings can either underregulate voltages or curtail excessive energy, reducing PV utilization [5]. Uniform curves applied across heterogeneous feeders often yield unequal curtailment, with customers farther from the transformer experiencing more frequent or deeper cuts, raising fairness and potential regulatory concerns [6]. Moreover, settings tuned offline for one scenario may perform poorly as topology, load mix, and PV profiles change over time. These limitations motivate advanced control designs that preserve the practicality of VW (local sensing, fast response) while optimizing the trade-off among voltage compliance, energy harvest, and fairness [7].

Several works have tackled overvoltage in PV-rich feeders using fixed or optimized VW designs. For example, [8] formulates a MINLP to tune VW parameters for maximizing DER injection while enforcing voltage limits and fairness. In [9], closed-form Volt-VAR-Watt settings are derived from voltage sensitivities, avoiding iterative optimization and communication while achieving performance comparable to optimized curves. The method in [10] designs smart-inverter VW parameters to mitigate violations and distribute curtailment more evenly, with simulation and hardware-in-the-loop validation. Complementary work in [11] analyzes VW stability limits and provides setting guidelines, while [12] uses a water-cycle algorithm to optimize VW settings for PV utilization and fairness. Coordinated voltage regulation using OLTCs and PV inverters has also been investigated in [13].

Although these VW designs offer clear advantages— such as simplicity, local implementability, and standards compliance, they remain intrinsically static and feeder-specific: parameters tuned offline for one topology, season, or PV mix can underperform as operating conditions drift [14]. Sensitivity-based settings rely on linearized models and typical cases, which may misestimate actions under nonlinearity or rare extremes; optimization-based designs can be computationally heavy, require re-tuning to track weather/load shifts, and still yield unequal curtailment across locations. Stability guidelines help avoid oscillations but typically rely on conservative settings that overlook energy efficiency and fairness [15].

In this paper, we propose a decentralized, data-driven MARL controller in which each PV inverter applies a shared policy to map local operating states to active-power setpoints in a supervisory VWC framework. Trained on diverse PV/load scenar-

ios, the policy adapts to changing operating conditions without manual re-tuning while balancing voltage compliance, PV utilization, and optional curtailment equity. A DNN-based safety layer filters provisional actions using a learned approximation of a voltage-feasibility correction. The resulting framework preserves local execution while improving voltage regulation and PV utilization under quasi-static operating conditions.

The contributions of this paper are as follows:

- A decentralized MARL-based supervisory Volt/Watt controller with shared policy execution at PV inverters.
- Data-driven adaptability across load and PV regimes without per-scenario or seasonal re-tuning.
- A lightweight DNN-based safety filter that approximates projection-based action correction and improves voltage-constraint satisfaction with millisecond-level inference.
- A reproducible benchmarking study on a realistic residential feeder using standardized voltage, curtailment, PV utilization, and fairness metrics.

The remainder of the paper is structured as follows. Section 2 describes the conventional and proposed model for VWC in ADNs. Section 3 presents the numerical results, which are based on real validated data. The paper concludes in Section 4.

2 Volt/Watt Control (VWC)

VWC keeps feeder voltages within operator-specified limits by curtailing DER active-power injection when local voltage exceeds predefined thresholds. Typically embedded in smart inverters, it provides autonomous voltage regulation without centralized coordination or extensive communication. However, poorly tuned VW settings can cause either unnecessary curtailment or insufficient over-voltage mitigation under high PV variability. The following subsections summarize conventional droop-based VWC and the proposed data-driven decentralized controller.

2.1 Conventional VWC

Traditionally, VWC is implemented using a canonical droop curve, as shown in Fig. 1. An inverter supplies its full available real power for voltages up to V_1; between V_1 and V_2, the output is reduced, typically linearly, as the local voltage rises; and at or above V_2, the active-power setpoint is driven to zero to prevent overvoltage. This piecewise profile, consistent with active-power control functions specified in interconnection standards such as IEEE Std 1547-2018 [4], provides a simple, fully local rule that limits voltage rise while curtailing generation only when necessary.

The main design task in conventional VWC is to select V_1, V_2, and the droop slope to minimize curtailment while keeping voltages within limits. These parameters are fixed before operation and are often applied uniformly across the feeder. Although this approach is simple and standards-friendly, it is feeder-specific and may become suboptimal as PV penetration, load composition, tap positions, or topology evolve. Fixed curves also ignore spatial voltage diversity, so electrically weak locations may experience more frequent curtailment. These limitations motivate adaptive VWC strategies that preserve local implementation while responding to changing operating conditions.

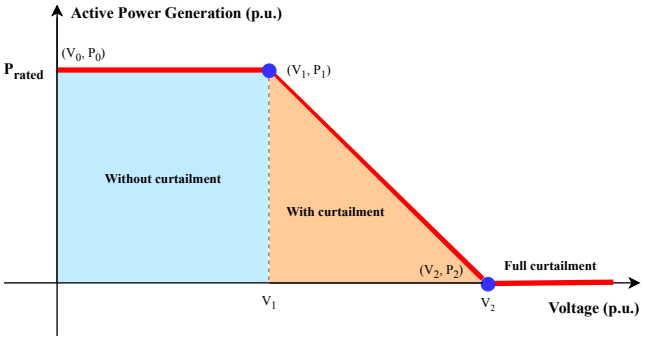


Figure 1: Standard VW characteristic.

2.2 VWC using MARL

To overcome the brittleness of fixed, feeder-specific VWC, we adopt a MARL paradigm in which all inverters share a learned policy that maps local/feeder states to active-power setpoints at each supervisory control interval. Training on diverse PV/load scenarios enables adaptation to nonlinear operating conditions without manual re-tuning [16]. At deployment, the policy uses local inputs and millisecond-level inference, preserving the simplicity of conventional V–W while removing centralized dependence and scaling naturally with the number of inverters [17]. By learning from data rather than fixed curves, the controller can exploit temporal and spatial operating patterns to increase PV utilization and reduce voltage excursions.

We cast VWC as a MARL problem in which each inverter is an agent that selects an active-power curtailment from local measurements. The controller is decentralized at execution yet data-driven: a single, shared policy is trained on diverse PV/load scenarios so it adapts to operating changes without manual re-tuning. Concretely, agent i observes a local vector $o_{i,t}$ (e.g., bus voltage magnitude, local load, and available PV) and outputs a curtailment $a_{i,t} \in [0, 1]$. The injected PV power is then

$$P_{i,t}^{\text{inj}} = \left(1 - a_{i,t}\right) P_{i,t}^{\text{avail}}. \tag{1}$$

Stacking all agents gives the joint action $\mathbf{a}_t = [a_{1,t}, \ldots, a_{K,t}]^\top$. Training follows the Centralized Training Decentralized Execution (CTDE) paradigm: critics see a global state s_t (concatenated or aggregated feeder features) to stabilize value estimation, while actors use only $o_{i,t}$ at run time.

We employ a parameter-shared, continuous-control actor–critic based on TD3 for MARL. Let $\pi_\theta(o_{i,t})$ be the deterministic actor and $Q_{\phi_j}(s_t, \mathbf{a}_t)$ the twin critics. Off-policy learning with replay buffers, target networks, and policy-smoothing noise improves sample efficiency and mitigates overestimation. The critic targets are

$$y_t = r_t + \gamma \min_{j=1,2} Q_{\bar{\phi}_j}(s_{t+1}, \tilde{\mathbf{a}}_{t+1}), \tilde{\mathbf{a}}_{t+1} = \pi_{\bar{\theta}}(\mathbf{o}_{t+1}) + \varepsilon, \tag{2}$$

with clipped noise ε and delayed policy updates. The actor is updated by the deterministic policy gradient

$$\nabla_\theta J(\theta) \approx \frac{1}{B} \sum_{t \in \mathcal{B}} \nabla_{\mathbf{a}} Q_{\phi_1}(s_t, \mathbf{a})\big|_{\mathbf{a}=\pi_\theta(\mathbf{o}_t)} \nabla_\theta \pi_\theta(\mathbf{o}_t) \tag{3}$$

where $\mathcal{B}$ is a mini-batch from the replay buffer.

The step reward balances voltage regulation and energy delivery:

$$r_t \;=\; -\alpha_v \sum_b \big[V_{b,t} - 1.05\big]_+ \;+\; \alpha_c \sum_i P^{\text{inj}}_{i,t}\, a_{i,t} \tag{4}$$

with $[\cdot]_+ = \max(0, \cdot)$. This design penalizes voltage excursions above 1.05 p.u. while encouraging PV energy delivery and limiting unnecessary curtailment. Because the policy is learned from data across many operating conditions, the resulting control is adaptive, avoids the one-size-fits-all limitations of fixed VW curves, and remains fully decentralized at deployment.

2.2.1 Fairness Beyond voltage compliance and energy harvest, curtailment equity among prosumers is important for customer acceptance and regulatory transparency. In feeders with spatial voltage diversity, static VW rules can impose disproportionate curtailment on electrically remote users. To capture this objective during learning, we optionally augment the base reward in (4) with an availability-weighted fairness regularizer that penalizes dispersion in curtailment across PV units.

Let $\mathbf{a}_t = [a_{1,t}, \ldots, a_{K,t}]^\top \in [0,1]^K$ be the curtailments at time t, and define the availability weights as

$$w_{i,t} \;=\; \frac{P^{\text{avail}}_{i,t}}{\sum_{j=1}^K P^{\text{avail}}_{j,t} + \varepsilon}, \qquad \bar{a}_{w,t} \;=\; \sum_{i=1}^K w_{i,t}\, a_{i,t}, \tag{5}$$

where $\varepsilon > 0$ is a small constant used to avoid division by zero when PV generation is zero. The weighted dispersion of curtailment is

$$F_t \;=\; \sum_{i=1}^K w_{i,t} \left(a_{i,t} - \bar{a}_{w,t}\right)^2, \tag{6}$$

which is zero when active PV units curtail equally in an availability-aware sense and increases with curtailment inequality. The fairness-augmented reward is then

$$r_t^{\text{fair}} \;=\; r_t^{\text{base}} \;-\; \alpha_f\, F_t, \tag{7}$$

where r_t^{base} denotes the reward in (4) and $\alpha_f \geq 0$ controls the equity–energy trade-off. Setting α_f=0 recovers the baseline reward, while larger values of α_f encourage more uniform curtailment at the possible cost of additional energy reduction.

For evaluation, curtailment equity is quantified using Jain's fairness index,

$$J = \frac{\left(\sum_{i=1}^K c_i\right)^2}{K \sum_{i=1}^K c_i^2}, \tag{8}$$

where c_i is the cumulative relative curtailment of PV unit i. The index satisfies $J \in [1/K, 1]$, with $J = 1$ indicating perfectly equal curtailment. This metric is distinct from the training regularizer F_t in (6).

2.2.2 DNN-based Safety Layer MARL does not strictly ensure constraint satisfaction at every control interval, especially under unseen PV/load conditions. Therefore, we place a learned action filter after the actor to correct potentially unsafe curtailment proposals. Given local observation $o_{i,t}$, available PV $P^{\text{avail}}_{i,t}$, and the actor's curtailment proposal $\hat{a}_{i,t} \in [0,1]$, the safety layer $\mathcal{S}_\psi$ outputs a corrected action

$$a^{\text{safe}}_{i,t} \;=\; \mathcal{S}_\psi\big(o_{i,t},\, \hat{a}_{i,t},\, P^{\text{avail}}_{i,t}\big) \;\in\; [0,1], \tag{9}$$

Algorithm 1: MARL training for VWC

Inputs: PV/Load dataset, feeder model; limits $V_{\min}$, $V_{\max}$, $S^{\max}$; shared actor π_θ, twin critics Q_{ϕ_1}, Q_{ϕ_2}; pretrained safety DNN $\mathcal{S}_\psi$; replay buffer $\mathcal{B}$.
Outputs: Safe curtailments $\mathbf{a}^{\text{safe}}$, injections $\mathbf{P}^{\text{inj}}$.
for each episode **do**
  Reset the environment to obtain the initial state s_t.
  for $t = 0, 1, \ldots, T-1$ **do**
    Propose: each agent i computes curtailment $\hat{a}_{i,t} = \pi_\theta(o_{i,t}) \in [0,1]$.
    Check: evaluate the joint proposal using power flow.
    if voltage and device limits are satisfied **then**
      $\mathbf{a}^{\text{safe}}_t \leftarrow \hat{\mathbf{a}}_t$.
    else
      $\mathbf{a}^{\text{safe}}_t \leftarrow \mathcal{S}_\psi(\mathbf{o}_t, \hat{\mathbf{a}}_t, \mathbf{P}^{\text{avail}}_t)$; clip to $[0,1]$.
    end if
    Execute: apply $\mathbf{a}^{\text{safe}}_t$; compute $\mathbf{P}^{\text{inj}}_t$; run PF; observe $\mathbf{o}_{t+1}$.
    Reward: compute r_t.
    Store: $\mathcal{B} \leftarrow \mathcal{B} \cup \{(\mathbf{o}_t, \hat{\mathbf{a}}_t, \mathbf{a}^{\text{safe}}_t, r_t, \mathbf{o}_{t+1})\}$.
    Update: periodically sample minibatches and perform TD3 critic, actor, and target-network updates.
  end for
end for

which is then applied to the inverter. The network $\mathcal{S}_\psi$ is trained to approximate the solution of a projection-based correction under a linearized feeder model. Denote the joint curtailment vector by $\mathbf{a} \in [0,1]^K$ and let v_0 be the pre-action voltage estimate at the controlled buses. Using a sensitivity matrix $S = \partial v / \partial P$ and the relation $P^{\text{inj}} = (\mathbf{1} - \mathbf{a}) \odot \mathbf{P}^{\text{avail}}$, the voltage change induced by curtailment can be approximated as

$$v(\mathbf{a}) \;\approx\; v_0 \;+\; H\,\mathbf{a}, \qquad H \;=\; -S\,\mathrm{diag}\big(\mathbf{P}^{\text{avail}}\big). \tag{10}$$

For each sampled operating state, the projection target is obtained by solving

$$\mathbf{a}^\star \;=\; \arg\min_{\mathbf{u}\in[0,1]^K} \;\big\|\mathbf{u} - \hat{\mathbf{a}}\big\|_2^2 \;\text{ s.t. }\; \underline{\mathbf{v}} \;\leq\; v_0 + H\,\mathbf{u} \;\leq\; \overline{\mathbf{v}}, \tag{11}$$

with tightened bounds $\underline{\mathbf{v}} = 0.95 + \varepsilon$ and $\overline{\mathbf{v}} = 1.05 - \varepsilon$ to provide a robustness margin. We generate an offline dataset by sampling diverse $(o, \hat{\mathbf{a}}, \mathbf{P}^{\text{avail}})$, computing v_0 and H from DistFlow/NR sensitivities, solving (11) to obtain $\mathbf{a}^\star$, and training $\mathcal{S}_\psi$ to minimize

$$\mathcal{L}(\psi) = \big\|\mathcal{S}_\psi(o, \hat{\mathbf{a}}, \mathbf{P}^{\text{avail}}) - \mathbf{a}^\star\big\|_2^2 + \beta\, \Phi\big(v_0 + H\,\mathcal{S}_\psi(\cdot)\big), \tag{12}$$

where $\Phi(\cdot)$ is a hinge barrier penalizing residual voltage-bound violations in the teacher model. At deployment, inference is millisecond-scale: each agent feeds its local features through $\mathcal{S}_\psi$, yielding $a^{\text{safe}}_{i,t}$. A final clipping step to $[0,1]$ is applied to respect the admissible action range. The learned filter is therefore interpreted as a fast constraint-aware correction, not as a formal safety guarantee. In the numerical experiments, it reduces unsafe actions and helps maintain voltage feasibility with negligible inference latency. Algorithm 1 summarizes the CTDE-TD3 training loop with on-demand DNN-based action filtering.

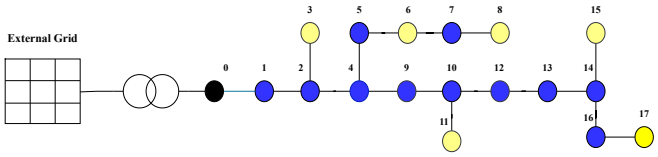


Figure 2: Modified CIGRE residential low-voltage feeder.

Table 1: Main hyperparameters for MARL training and DNN safety layer

Hyperparameter	Value
MARL training	
Learning rate	1×10^{-4} actor/critic
Discount factor γ	0.995
Batch / buffer size	512 / 5×10^5
Target update τ	1×10^{-2}
Policy noise	$\mathcal{N}(0, 0.1)$, clip ± 0.5
Exploration noise	Std 0.3, decay 0.999 $\rightarrow$ 0.2
Episodes / steps	1000/96
Action / voltage limits	[0, 1], [0.95, 1.05] p.u.
DNN safety layer	
Inputs	$(o_i, \hat{a}_i, P_i^{\text{avail}})$
Output range	Clipped to [0, 1]
Learning rate / weight decay	10^{-3} / 10^{-5}
Batch size / epochs	512/50
Train/validation split	80%/20%
Runtime target	$<$ 1 ms per call

3 Numerical Analysis

3.1 Dataset and training setup

We evaluate the proposed controller on a modified CIGRE residential low-voltage feeder [18] with 18 buses (single-ended radial), shown in Fig. 2. Bus 0 is the slack; the remaining nodes supply aggregated residential loads. Six PV inverters are installed at buses {3, 6, 8, 11, 15, 17} (increasing electrical distance from the transformer), each controlled by one agent under the decentralized MARL scheme. Time series are sampled at 15-min resolution. Load profiles are standard residential demand traces bundled with the benchmark feeder and scaled to MW; nonphysical negatives are clipped to zero.

PV availability profiles are derived from historical PV production data published by the Elia Group (Belgian TSO) and resampled to 15-min to match the loads [19]. These traces are mapped to the six inverters according to their nameplates to obtain $P_i^{\text{avail}}(t)$. All experiments enforce typical LV voltage bounds $V_{\min} = 0.95$ p.u. and $V_{\max} = 1.05$ p.u., inverter apparent-power limits $S_i^{\max}$, and standard AC power-flow feasibility. Training uses a year-long pool of PV/load scenarios; held-out day-level episodes are used for testing and reporting.

Table 1 lists the main MARL and DNN safety-layer hyperparameters. Small learning rates, large batches, delayed policy updates, and soft target updates are used to stabilize TD3 training. Each episode represents one day with 96 time steps. The safety DNN is kept lightweight to allow millisecond-level inference during deployment. A 1,000-episode training run completed in approximately 15 minutes on a workstation-class laptop.

3.2 Results

In this section, the proposed MARL strategy is implemented on the 18-bus active distribution network to evaluate its ability to improve voltage profiles while minimizing PV curtailment. We

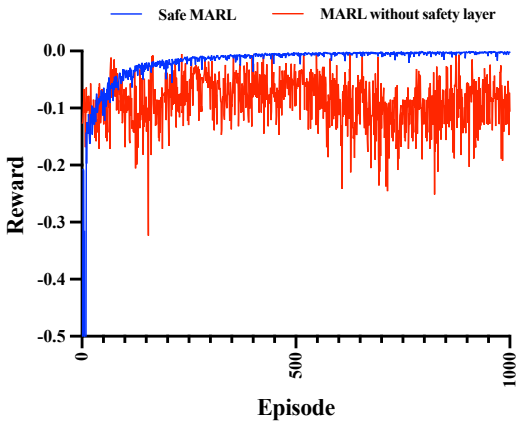


Figure 3: Convergence of the proposed approach.

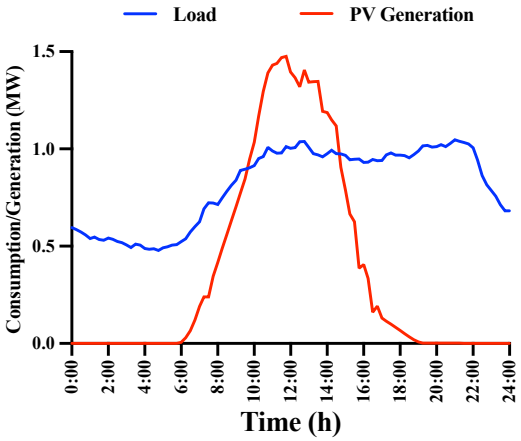
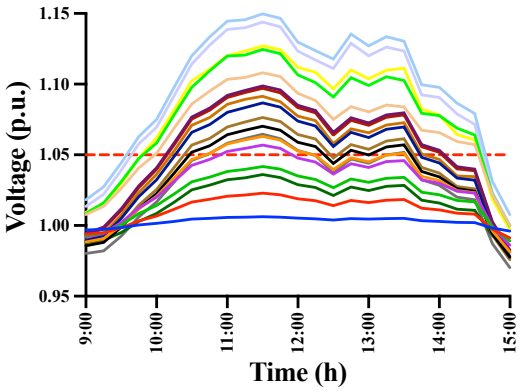


Figure 4: Load and PV generation profiles of the test day.

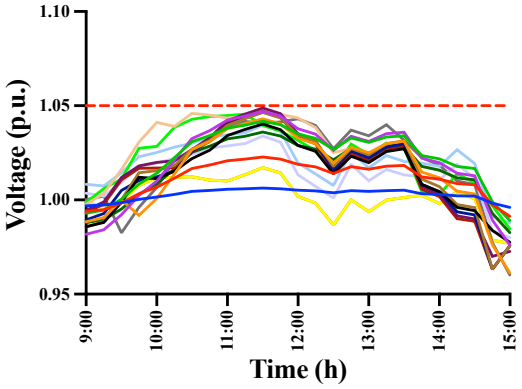
first compare the training convergence of the MARL controller with and without the DNN safety layer. As shown in Fig. 3, adding the safety layer accelerates learning, improves the final return, and reduces training variance. By correcting potentially unsafe actions before execution, the safety layer reduces penalty spikes and provides cleaner replay data for critic updates. Without this correction, frequent voltage violations slow convergence and increase return variability. Overall, the safety layer improves training stability and leads to policies that reduce voltage-violation magnitude and duration while avoiding excessive curtailment.

Having established the training stability and benefit of the safety layer, we now turn to out-of-sample evaluation on a held-out test day. Figure 4 shows aggregate PV generation versus total load. Between roughly 09:00–15:00, PV output exceeds demand, creating reverse-power-flow conditions. In the no-control baseline, this produces widespread over-voltage, with several buses rising above the 1.05 p.u. limit, as shown in Fig. 5(a). With the proposed MARL controller, voltages remain within the admissible band and the 1.05 p.u. cap is respected throughout the high-PV period, as shown in Fig. 5(b). This is achieved through time-aligned active-power curtailment of the PV units during midday peaks, as shown in Fig. 6, eliminating over-voltage while preserving generation outside congested hours.

Beyond voltage compliance and energy yield, the distribution of curtailment across prosumers is also important because persistent location-dependent curtailment can penalize specific users,



(a) Base Case



(b) MARL

Figure 5: Voltage profiles of all buses on a held-out test day.

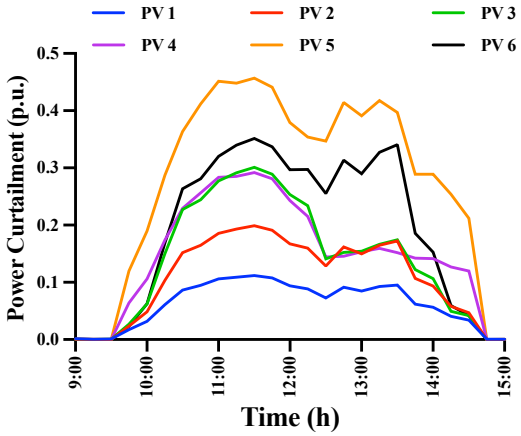


Figure 6: Power curtailments of PVs.

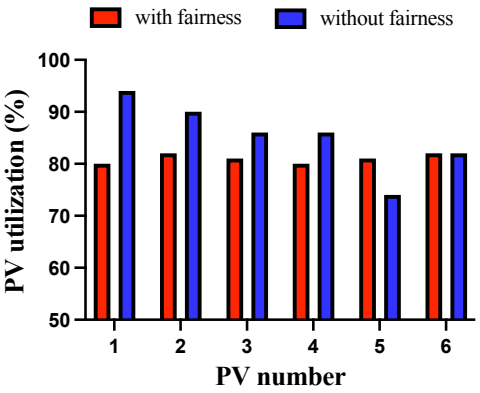


Figure 7: PVs utilization without and with fairness.

particularly those electrically far from the transformer. We therefore assess how the proposed controller allocates curtailment among PV units relative to the unregularized baseline. Figure 7 compares the per-PV energy utilization with and without fairness regularization. With fairness enabled, the dispersion across units decreases and the bars become nearly uniform, indicating that curtailment is shared more evenly across inverters. This equity improvement comes with a modest efficiency cost: overall PV utilization decreases from 85.7% without fairness to 81.3% with fairness, because some inverters curtail slightly more than strictly necessary to equalize the burden. In practice, the fairness weight provides a tunable parameter that allows operators to trade a few percentage points of energy for improved curtailment equity while maintaining voltage limits.

Beyond the single-day analysis, Table 2 compares month-long performance. Without fairness, the proposed MARL controller records no voltage violations in the tested scenarios, whereas the fixed VW baselines based on [10,12] still incur violation rates of 0.67% and 0.39%, respectively. The peak voltage is also lower with MARL, reaching 1.047 p.u. compared with 1.058 p.u. and 1.054 p.u. for the two fixed VW baselines. At the same time, PV utilization is higher, reaching 84.87% compared with 71.45% and 73.47%. These differences arise from the one-size-fits-all nature of curve-based designs: a single (V_1, V_2, slope) tuned offline cannot fully track hour-to-hour and bus-to-bus variability in PV generation, load, and feeder sensitivity. In contrast, the proposed policy adapts its curtailment decisions to the observed operating state at each supervisory control interval, while the DNN safety layer filters potentially unsafe proposals before execution.

Table 2: Performance over one month with and without fairness

Metric	Base case	[10]	[12]	Our Approach
Without fairness				
Violation rate (%)	3.58	0.67	0.39	0.00
Max voltage (p.u.)	1.162	1.058	1.054	1.047
PV utilization (%)	100	71.45	73.47	84.87
Curtailment equity	-	0.83	0.77	0.81
With fairness				
Violation rate (%)	3.58	0.72	0.45	0.00
Max voltage (p.u.)	1.162	1.06	1.055	1.048
PV utilization (%)	100	64.85	67.47	79.75
Curtailment equity	-	1	1	0.98

With fairness enabled, the fixed designs in [10, 12] still exhibit violation rates of 0.72% and 0.45% and lower PV utilization of 64.85% and 67.47%, respectively. In contrast, the proposed method maintains zero observed violations in the tested scenarios and achieves 79.75% PV utilization. The fixed designs attain a curtailment equity index of 1.00, while the proposed method reaches 0.98. Thus, the proposed method does not maximize the equity index itself, but maintains near-uniform curtailment while allowing spatially differentiated actions when required by the grid. The fairness term acts as a soft, availability-weighted penalty rather than enforcing identical curtailment across all units. This retains higher PV utilization and voltage compliance at a small reduction in the equity index in the tested scenarios.

4 Conclusion

This paper presented a decentralized supervisory Volt/Watt control (VWC) scheme for PV-rich active distribution networks (ADNs) based on Multi-agent reinforcement learning (MARL) with an on-demand DNN safety layer. The MARL policy learns state-feedback curtailment from diverse PV/load scenarios and executes locally with millisecond-level inference, while the safety layer filters potentially unsafe proposals using a learned voltage-feasibility correction. On a modified 18-bus residential feeder, the proposed controller eliminated observed voltage violations in month-long out-of-sample tests, reduced peak voltages, and increased injected PV energy compared with fixed curve designs. With the fairness term enabled, curtailment became nearly uniform across inverters with only a modest reduction in utilization.

Overall, the results show that a learned state-feedback policy can overcome the one-size-fits-all limitations of static Volt/Watt curves, while a lightweight safety filter improves constraint satisfaction without centralized online optimization. Future work will extend the framework to joint Volt–Var–Watt control, coordination with OLTCs, robustness under unseen operating conditions, and validation on real feeders and hardware-in-the-loop platforms.

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